```
import pandas as pd

df = pd.read_csv('/content/creditcard.csv')
```

```
df.head()
```

	Time	V1	V2	V3	V4	V5	V6	V7	V8	V9	...	V21	V22
0	0	-1.359807	-0.072781	2.536347	1.378155	-0.338321	0.462388	0.239599	0.098698	0.363787	...	-0.018307	0.277838
1	0	1.191857	0.266151	0.166480	0.448154	0.060018	-0.082361	-0.078803	0.085102	-0.255425	...	-0.225775	-0.638672
2	1	-1.358354	-1.340163	1.773209	0.379780	-0.503198	1.800499	0.791461	0.247676	-1.514654	...	0.247998	0.771679
3	1	-0.966272	-0.185226	1.792993	-0.863291	-0.010309	1.247203	0.237609	0.377436	-1.387024	...	-0.108300	0.005274
4	2	-1.158233	0.877737	1.548718	0.403034	-0.407193	0.095921	0.592941	-0.270533	0.817739	...	-0.009431	0.798278

5 rows × 31 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 61491 entries, 0 to 61490
Data columns (total 31 columns):
 #   Column  Non-Null Count  Dtype  
---  ------  --------------  -----  
 0   Time    61491 non-null  int64  
 1   V1      61491 non-null  float64
 2   V2      61491 non-null  float64
 3   V3      61491 non-null  float64
 4   V4      61491 non-null  float64
 5   V5      61491 non-null  float64
 6   V6      61491 non-null  float64
 7   V7      61491 non-null  float64
 8   V8      61491 non-null  float64
 9   V9      61491 non-null  float64
 10  V10     61491 non-null  float64
 11  V11     61491 non-null  float64
 12  V12     61491 non-null  float64
 13  V13     61491 non-null  float64
 14  V14     61491 non-null  float64
 15  V15     61491 non-null  float64
 16  V16     61491 non-null  float64
 17  V17     61491 non-null  float64
 18  V18     61491 non-null  float64
 19  V19     61491 non-null  float64
 20  V20     61491 non-null  float64
 21  V21     61491 non-null  float64
 22  V22     61491 non-null  float64
 23  V23     61491 non-null  float64
 24  V24     61491 non-null  float64
 25  V25     61491 non-null  float64
 26  V26     61491 non-null  float64
 27  V27     61491 non-null  float64
 28  V28     61491 non-null  float64
 29  Amount  61490 non-null  float64
 30  Class   61490 non-null  float64
dtypes: float64(30), int64(1)
memory usage: 14.5 MB
```

```
df['Class'].value_counts()
```

	count
Class	
0.0	61327
1.0	163

dtype: int64

```
total = len(df)
fraud = df['Class'].sum()

percentage = (fraud / total) * 100
print(percentage)
```

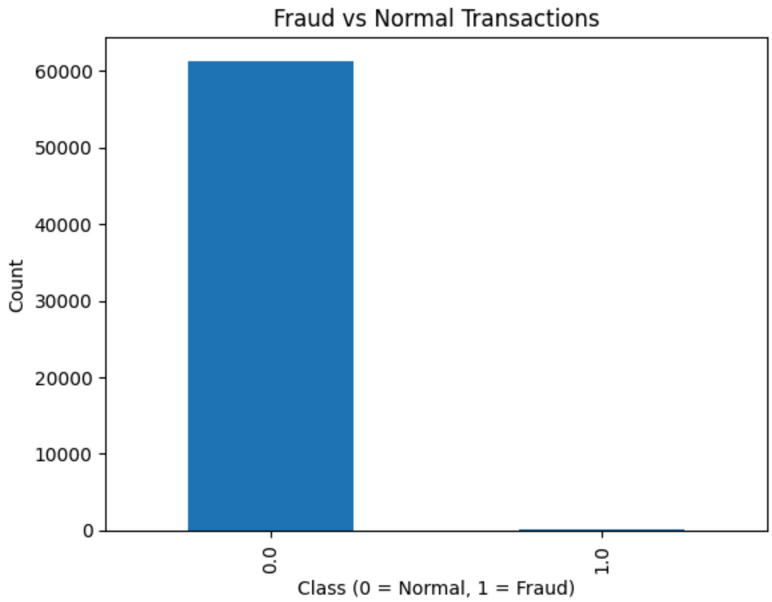
```
0.26507944252004356
```

```
print("Fraud Percentage:", round(percentage, 2), "%")
```

```
Fraud Percentage: 0.27 %
```

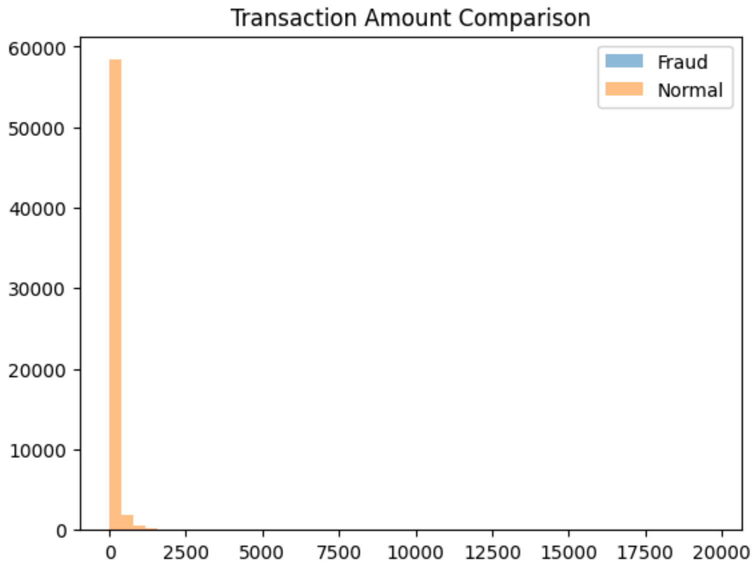
```
import matplotlib.pyplot as plt

df['Class'].value_counts().plot(kind='bar')
plt.title("Fraud vs Normal Transactions")
plt.xlabel("Class (0 = Normal, 1 = Fraud)")
plt.ylabel("Count")
plt.show()
```

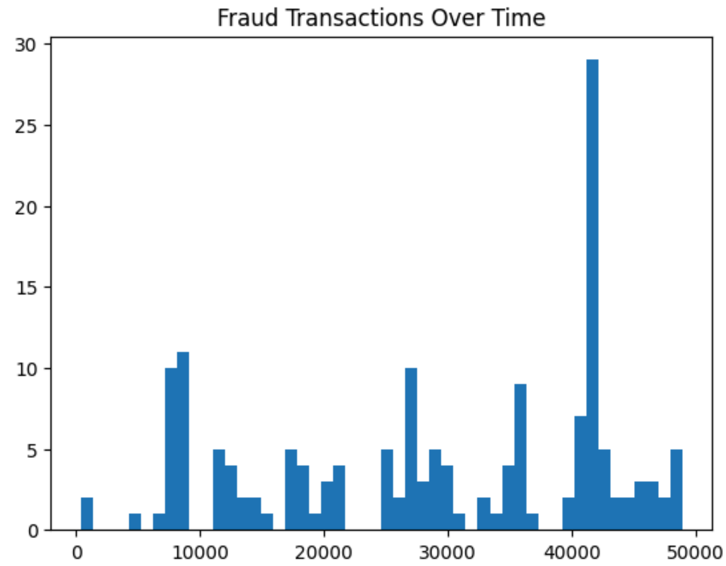



```
fraud = df[df['Class'] == 1]
normal = df[df['Class'] == 0]

plt.hist(fraud['Amount'], bins=50, alpha=0.5, label='Fraud')
plt.hist(normal['Amount'], bins=50, alpha=0.5, label='Normal')
plt.legend()
plt.title("Transaction Amount Comparison")
plt.show()
```




```
plt.hist(fraud['Time'], bins=50)
plt.title("Fraud Transactions Over Time")
plt.show()
```




```
fraud = df[df['Class'] == 1]
normal = df[df['Class'] == 0]

print("Fraud Amount Avg:", fraud['Amount'].mean())
print("Normal Amount Avg:", normal['Amount'].mean())
```

```
Fraud Amount Avg: 93.11300613496931
Normal Amount Avg: 95.44060707355652
```

```
print("Fraud Max:", fraud['Amount'].max())
print("Fraud Min:", fraud['Amount'].min())
```

```
Fraud Max: 1809.68
Fraud Min: 0.0
```

```
high_fraud = fraud[fraud['Amount'] > 1000]
print(len(high_fraud))
```

```
2
```